Conceptual design of a medical drone logistics network for Scotland

B. Jones[a,d], G. Filippi[a], T. Basu[c], B. Parsonage[a], E. Patelli[b], C. Maddock[a], M. Vasile[a], M. Fossati[a]

[a] *Mechanical and Aerospace Engineering, University of Strathclyde, Glasgow G1 1XQ, UK*
[b] *Civil and Environmental Engineering, University of Strathclyde, Glasgow G1 1XQ, UK*
[c] *School of Mathematics, Statistics and Physics, Newcastle University, Newcastle NE1 7RU, UK*
[d] *Corresponding author, bryn.jones@strath.ac.uk*

Abstract

Distributed health care networks have been a key area of interest in improving the health care systems and with the recent developments of cost-effective unmanned autonomous vehicles, the use of drone for health care delivery is also gaining interest from different government organisations. Moving towards this direction, NHS (National Health Service) Scotland is looking into the prospect of using drones for medical deliveries across Scotland. This work illustrates the design of an optimal network for this use case introducing digital twinning at different levels of the delivery network. To design such networks, different quantities of interest are be considered such as flight time, capital expenditure, and risk of impact. A specific focus on multi-objective trade-off concepts for network design is used that demonstrates how different network solutions affect the delivery of potentially life-saving medical goods.

A set of random network failures are considered and the efficiency of the network after adapting to the failures is calculated as a measure of the network resilience. Network efficiency is a function of the expected delivery time by drones using optimal paths. A Pareto front of solutions is generated, and selected ones are plotted in terms of physical location, infrastructure, and nominal delivery routes. The results show that investing in more charge stations across Scotland reduces the overall delivery times by reducing reliance on fixed-wing deliveries, whereas investing in more drone ports increases the network resilience by reducing the increased delivery burden on remaining drones when a drone port fails.

Keywords: Drone Logistic Network, Emergency Response Service, Resilience Engineering, Multi-objective optimisation

1. Introduction

In recent years, the effectiveness of healthcare logistics via a drone-based delivery network has been subject of significant study. The transport of microbiological specimens including blood cultures by drone in Africa was studied in [1]. Successful flight tests for medical delivery have been conducted in Spain[2], in the US, Canada and Italy. An additional study on the impact of drone transportation on biological samples was explored in [3], revealing no adverse effects for turnaround times of less than 4 hours. Feasibility of drone logistic networks for delivering medical goods was observed by Leonardo and Telespazio [4] near Rome, Italy and by Matternet [5] in Berlin, Germany. Similar initiatives were undertaken in Switzerland with Swiss Post, involving the transportation of laboratory samples between two hospitals [6].

In the UK specifically, a number of studies have been undertaken looking into the feasibility of medical drone logistics networks. In Southampton, Grote et al. [7] and Oakey et al. [8] looked into the serviceability of drones to various GP surgeries from the main hospital in the area, as well as the cost of replacing the existing road vehicle deliveries with drone-based ones, finding that the delivery times would be cut by 72% at the cost of a 56% increase in spending. Since service requirements were already being met, they estimated that the cost of the drones would need to drop to 19% of current values to be financially viable. Porter et al. [9] expanded the case study to Dorset, finding similar results of an 83% reduction in delivery times coupled with a 58% increase in costs. Though, the studies stress that the key advantage of the drone-based deliveries are reductions in CO2 emissions as well as more equitable service to remote rural communities.

Some feasibility studies were also conducted for a potential medical drone logistic network in London. Arenzana et al [10] find that a drone logistic network could have approximately 30% of the operational costs of the existing ground-vehicle based network, with a 90% reduction in energy/fuel costs, due to London's congested road network. A white paper by Nesta [11] shows how drones can be used to reduce delivery times and costs, while also identifying a number of challenges that need to be overcome such as airspace management and beyond visual line of sight operations.

A randomised controlled laboratory study by Wiltshire et al. in Northumbria found no adverse effects in blood sample quality when delivered by drone compared to ground vehicle [12].

The UK government is exploring the possibility of an autonomous drone logistic network to facilitate the delivery of medical equipment and assistance to remote areas. The first step in this process is the network design, which is the focus of this paper, with a specific application to Scotland.

Within the context of Scotland, there are a great number of design challenges that could be considered. There are hundreds of locations to connect to the network, many of which are not within a typical vertical take-off and landing (VTOL) drone's range of the main cluster. There are also vast areas which may be put under flight restriction to accommodate military training exercises, as well as the typical controlled airspaces around airfields, and a few forbidden zones around nuclear facilities; in addition to this, arbitrary zones can be defined on a case-by-case basis with suitable forewarning. The weather conditions are notoriously unpredictable and adverse, which is particularly challenging for drones.

Despite the challenges, there are also significant opportunities offered uniquely by drones, especially in improving the equity of care to the remote communities and the efficiency of servicing them. For example, medical supplies or test samples delivered to or from island communities currently rely on ferries, a relatively slow service with a fixed schedule and little redundancy. Similarly, remote mainland communities are serviced via slow circuitous roads, making regular pick-ups expensive relative to the number of items to be delivered. The use of drones addresses all of these problems, being able to fly faster and in more direct paths, and potentially being more cost effective at smaller scales.

To address the design challenge, it is necessary to establish a way of evaluating different network designs. To achieve this, each drone network design is considered as a vehicle routing problem (VRP) to find the optimal routes and schedules for the drones for a given set of delivery missions, then the various delivery times are collated to calculate the network performance metrics. Research in the area of logistics algorithms and solutions for the VRP is increasingly considering the use of drones as their technology continues to develop [13] and allows for increasingly more reliable delivery.

The delivery strategy and optimisation algorithm applied largely depends on the context of the problem, for example Pachayappan and Sundarakani develop and compare flexible and non-flexible strategies for managing drone battery charge for sequences of deliveries in a hyperlocal market [14]. Saleu et al. pose the problem of deliveries using multiple road vehicles which deploy drones and develop a heuristic method for solving it [15]. Choudhury et al. develop a computationally efficient algorithm for resolving a drone logistic network, and demonstrate its use for a network of drones which piggyback on public transport to conserve energy.

In terms of network design, Pinto and Lagorio offer a particularly relevant study to the current use case [16], which involves the use of a hub-and-spoke style architecture with intermediate charging stations to cover delivery points outside of the drone's range. They also employ a shortest path based heuristic which improves the computational cost of their bi-objective optimisation with near-zero sacrificed in terms of the optimality of the solution. Cokyasar et al. develop a method for drone network design with automated battery swapping stations [17], including a queueing system for the battery swaps that accounts for the network congestion. One of their findings is that drone deliveries can be more optimal than trucks in terms of cost, even when several stops for swapping batteries are required along the routes. They also find that queue times for battery swaps have a significant impact on the performance of the network. In the current paper, queues for both battery swaps and take-off/landing pads are considered, as well as several heuristics for the VRP algorithm that keep the computational cost tractable considering the outer loop of multi-objective design optimisation.

This paper demonstrates a multi-objective network design optimisation for a medical goods drone network in Scotland. A Pareto front of solutions is generated, which can inform executive decisions in terms of trade-offs between performance and cost metrics for the construction of the actual physical network. The design challenge takes into account factors like capital and operational costs, delivery time, network reliability, and resilience to unforeseen events. This paper further advances what was proposed by the authors in [18, 19, 20].

The following section describes in more detail how the network is modelled, followed by mathematical descriptions of the resilience and optimisation problem in sections 3 and 4. Then, the specific application to a medical goods drone network in Scotland is described, including the results of the optimisation, finally leading to the conclusions section where the main findings of the paper are summarised.

2. Modelling

2.1. Network representation

The network is represented as a graph with set of fixed nodes, which may be active or inactive depending on the infrastructure assigned to each of them for a given design. Edges between each pair of nodes are pre-computed for each drone type considered, accounting for the actual path of the drone, and are deactivated if the path is longer than the drone's range. The weight of each edge is a combination of fixed components: take-off time, flight time, landing time, and battery swap time; and a variable component of queue time for: take-off, landing, and battery swap; the variable component of which depends on the state (congestion) of the network at particular points in time.

Figure 1 shows different NHS locations and laboratories; the objective is to find an optimal design to connect these locations as well as obtain a path for delivery as explained in section 2.4.

The NHS locations are associated with one of the so-called 'health boards' within Scotland; for the purposes of modelling the network in a realistic way, each board is represented as a hub-and-spoke model, with one or more hospitals in each health board designated as a hub. Each of the hubs in the national network are then represented as a point-to-point network. An example for two of the health boards is shown in figure 2. A more detailed explanation of how the graph is implemented is found in section 4.3.

2.2. Flying constraints

The design and performance of a delivery network of drones needs to take into account flying restrictions such as no-fly zones, wind speed, temperature and population density.

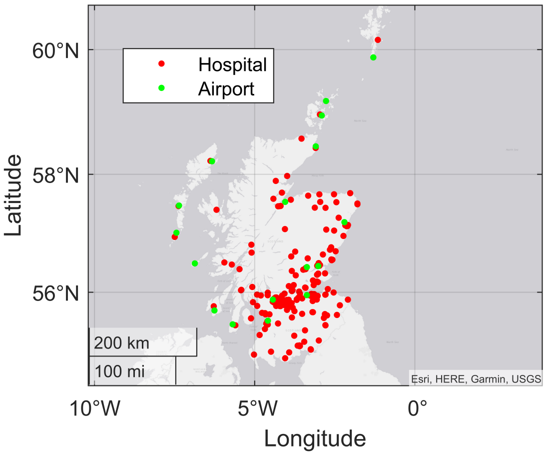

Figure 1: NHS locations and commercial airfields within Scotland.

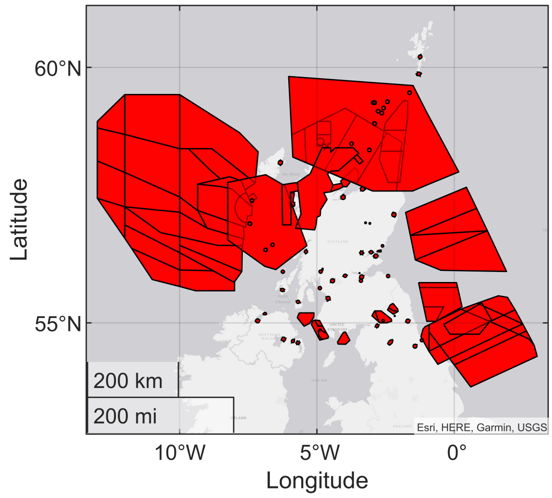

Figure 3: All standard restricted or forbidden airspaces in Scotland (not to be used as a reference for flight planning). Additional arbitrary zones can be issued on a temporary basis, typically to ensure safety around military training exercises.

Figure 2: Hub-and-spoke representation of Borders and Dumfries & Galloway health boards.

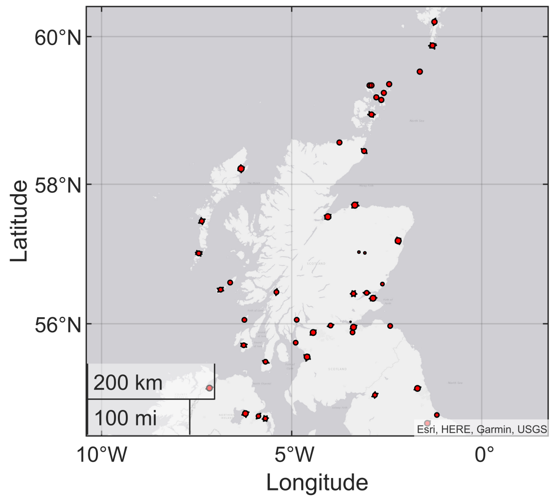

Figure 4: All permanently restricted or forbidden airspaces in Scotland (not to be used as a reference for flight planning).

No-fly zones are due to critical infrastructures or areas for which drones could represent or be subjected to risk and safety issues. These areas are then interdicted to potential flight paths. Different types of no-fly zones are possible and the main categories are restricted, danger, prohibited areas. A representation of No-Fly Zones in Scotland for all the three categories is shown in fig. 3 where the areas are defined as a list of 2-D polygons. Some of these areas are only temporarily activated during military training exercises, for the purposes of this paper only permanent restriction/prohibited zones are considered, shown in fig. 4.

Most of these permanent flight restriction zones are for airfields. Except for military airfields which are always considered prohibited, a flight path that intersects an airfield's flight zone is only considered valid if the source or destination of the path lies within that airfield's flight zone. Otherwise, any path that intersects with any of these zones is considered invalid. This is to minimise scheduling conflicts and disruption with these airfields, since the drones typically fly below the ceiling of these zones.

In the actual operation of the physical network, the no-fly zones would need to be monitored on a continual basis, since changes may occur at any time, and additional zones can be defined arbitrarily according to requirements from other users of the airspace. All of this would need to be taken into account to efficiently manage the network and to submit valid flight plans.

2.3. *Payload thermal model*

Temperature control of the payload is essential to satisfy thermal constraints on the medical items. The temperature control system considered is a passive system that includes an insulating package, a medical payload and a Phase Change Material (PCM). A lumped mass model including thermal conduction and convection have been developed as shown in Figure 5 and verified against the results from ATMOS high fidelity software from Intelsius (see fig. 6).

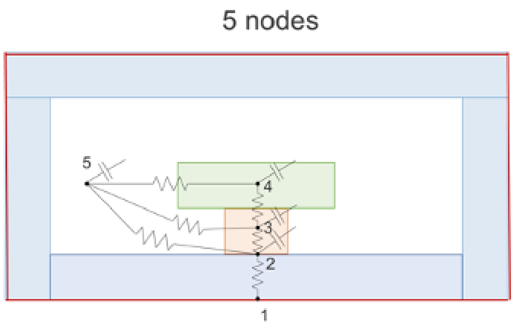

Figure 5: Lumped mass model for the medical thermal system.

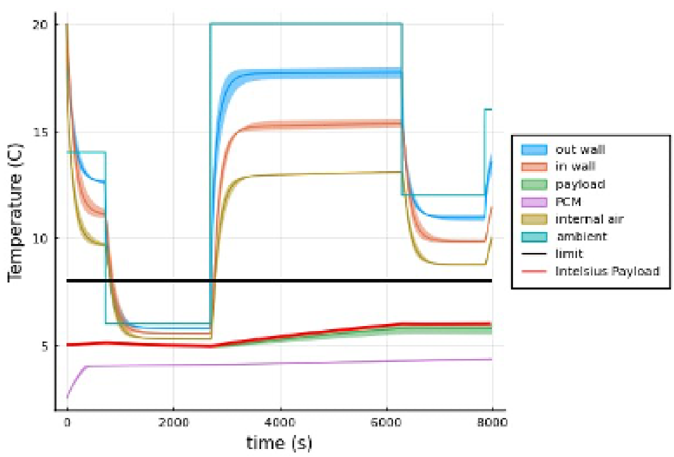

Figure 6: Verification of the thermal model.

The results from this thermal model give an upper bound on the duration of a given delivery mission to ensure that it stays within the appropriate temperature range throughout the mission.

2.4. *Individual path planning*

A further problem considered as part of the network design is path planning. This is not an independent problem since it is coupled with delivery time (payload temperature, flight range) and ground impact risk. Population density is an essential factor to be considered since it affects the risk of ground impact. In traditional aviation, the primary safety considerations refer to the lives and property of the crew and passengers. In contrast, the foremost safety apprehension in Unmanned Aircraft Systems (UAS) operations is the risk encompassing individuals and properties not associated with the UAS operation. New risk indicators are then under research and development in [21]. Given a point to point flight connection, the optimal path that connects them is found considering minimum time and risk objectives, and the constraint of the drone range given the wind velocity, although in this paper wind velocity is kept constant. The space around the pickup and drop-off locations is initialised as a regular two-dimensional grid with population density and no-fly zone data. This is transformed into a graph by considering grid points as nodes, with orthogonally and diagonally adjacent grid points connected via undirected edges, unless they intersect a no-fly zone. The edge weights are a linear combination of the population density and the distance between the two grid points. Using a standard graph search algorithm such as Dijkstra's, various paths balancing time and risk are found by varying the relative proportional linear combination of these factors in the edge weights. Figures 7 and 8 show the results for the bi-objective path search problem where the objectives are time and risk of ground impact.

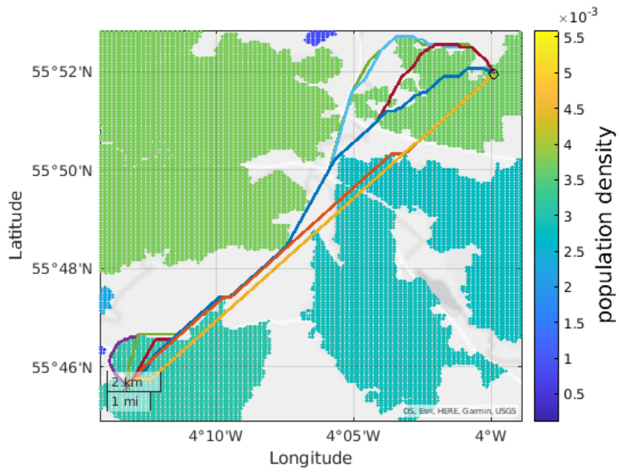

Figure 7: Path planning w.r.t. risk and time

3. Network Resilience

Resilience is an important factor for the Scottish medical drone network concept. Many of the packages will be both unique and time-limited, and there is the potential that the network could be used for missions which are life-saving and time-critical. In addition, as small air vehicles, the drones are more vulnerable to disruption than most other delivery methods. Therefore, a resilience model is necessary to properly evaluate and rank network designs against one another. This section describes the way that the resilience is modelled in this work.

The resilience of a complex system is a property related to the dynamical capability to react and adapt to external and/or internal unpredicted events in order to maintain its functionality. This definition implies the ability to absorb shocks due to

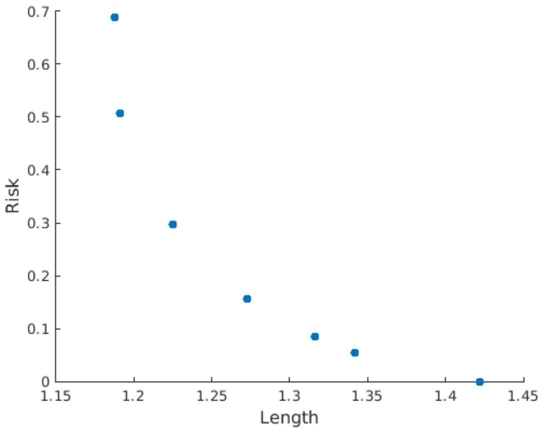

Figure 8: Path planning with bi-objective Pareto solutions

uncertain events, minimising the loss of performance, and to recover partially or entirely to the previous nominal conditions.

Consider a generic network configuration defined by a set of stations, infrastructures attached to the stations, and a heterogeneous fleet of available drones. The information about this network configuration is condensed in the vector of design decision variables $\mathbf{d} \in D$ and in the vector of uncertain variables $\mathbf{u} \in U$.

The network resilience proposed here is calculated as the expected value of an operator Φ:

$$\text{Resilience} = \mathbb{E}\left(\Phi_\lambda\left(\mathbf{d}, \mathbf{u}_\lambda\right)\right) = \sum_{\lambda\in\Lambda} p_\lambda \Phi_\lambda\left(\mathbf{d}, \mathbf{u}_\lambda\right) \tag{1}$$

where p_e is the the probability of the event $\lambda \in \Lambda$ and the operator Φ_λ models the network dynamics under the uncertain event λ. Following (1) there are two objects that have to be defined: the probability space and the operator Φ.

3.1. Probability space

We consider a set of independent events λ_i that includes the functional state of each node in the network (totally functioning or totally failed) and the flight range variation of each drone type due to weather conditions.

To cope with imprecision and lack of knowledge about the frequency of these events, the probability of them to happen is assumed to be elicited from expert opinion, giving rise to probability bounds $p_{\lambda_i} = [\underline{p}_{\lambda_i} \bar{p}_{\lambda_i}]^T\ \forall \lambda_i \in \Lambda_i$.

Since the elementary events are assumed to be mutually independent and the probabilities of the elementary events λ_i to be in the interval $[0, 1]$, the probability space of events $\lambda \in \Lambda$ is given by the Cartesian product of the elementary p-boxes:

$$p_\lambda = \left[\prod_i \underline{p}_{\lambda_i}, \prod_i \bar{p}_{\lambda_i}\right]^T. \tag{2}$$

3.2. Network dynamics

The second element in (1) is the operator Φ_λ which quantifies the ability of the network to minimise the loss of performance due to the uncertain event, and to recover partially or fully after it. The operator Φ is related to the network efficiency ϵ which models degradation and recovery under uncertain external/internal events as function of time for delivery.

In nominal condition, the delivery time of mission m between source (pick-up) node γ_m^p and sink (delivery) node γ_m^d is:

$$T_{nom}^m = \min_{\mathbf{p}} \left[\mathbb{E}\left(\text{time}_{\text{nom}}^m\left(G, \mathbf{p}, \mathbf{u}\right)\right)\right], \quad \forall \gamma \in \Gamma \tag{3}$$

where $\mathbf{p}$ is the part of the decision vector $\mathbf{d}$ that refers to the route plan definition, and G is the network graph.

The nominal performance indicator, the nominal network efficiency, is the constant function over time:

$$\epsilon_{nom} = \sum_{m=1}^{M} \frac{1}{T_{nom}^m} \tag{4}$$

The realisation of a generic event $\lambda \in \Lambda$ modifies the topology of the network by deactivating one or more stations and/or reduces the nominal distance a drone can fly. After event λ then, the delivery time of mission m between source node γ_m^p and sink node γ_m^d becomes:

$$T_\lambda^m(t) = \min_{\mathbf{p}} \left[\mathbb{E}\left(\text{time}_{\lambda,t}^m(G, \mathbf{p}, \mathbf{u})\right)\right] \tag{5}$$

In (5) it is considered the possibility for failed nodes in the network to recover as consequence of targeted maintenance. In particular, a policy is included in the model for the decision of the sequence of nodes to re-activate.

Similarly to the nominal case, the network efficiency after event λ is:

$$\epsilon_\lambda(t) = \sum_{m=1}^{M} \frac{1}{T_\lambda^m(t)} \tag{6}$$

In this case, due to the repairing policy, $\epsilon_\lambda(t)$ is a monotonic function increasing over time.

The network efficiency in (6) is then normalised based on the nominal condition:

$$\hat{\epsilon}_\lambda(t) = 1 - \frac{\epsilon_{nom} - \epsilon_\lambda(t)}{\epsilon_{nom}} \tag{7}$$

4. Optimisation

The Scottish medical goods drone network, from a conceptual point of view, is on the scale of potentially hundreds of stations, of which thousands of different designs should be evaluated to identify a Pareto front of solutions, where each design evaluation also should consider an array of failure cases to model the resilience. This means that considering the full NP-hard logistical problem is not feasible in terms of computational cost.

Instead, the network evaluation is set up in a way that it can be solved with a modified fastest path graph search algorithm, with dynamic edge weights accounting for the congestion of the network, and multiple layers accounting for the two drone types. This also has the benefit of being directly applicable as a tactical network operations algorithm, where the planning and scheduling of the network can be quickly evaluated in real-time response to failure events.

The basic optimisation problem considered is the vehicle routing problem. Because the Scottish health network is distributed over a large area, both multiple-depot and heterogeneous-fleet variations are considered. Enabling the use of multiple sparsely packed depots allows localised deliveries to be much more efficient. Similarly, allowing for a mix of short-range and long-range drones allows for more specialised drones for each mission, improving efficiency, but also the resilience of the network is improved via non-overlapping failure modes.

Because of the nature of the use case, there are a fixed number of pick-up and drop-off locations, corresponding to various labs, health centres, hospitals, etc. in Scotland, although there could be multiple missions scheduled between the same two locations. However, some locations are naturally more active than others, e.g. major hospitals, so the simulation of the network in terms of scheduled missions follows a hub-and-spoke structure, with multiple hubs distributed across Scotland corresponding to the various health board regions.

The remainder of this section describes in more detail and in mathematically formal description the optimisation problem set-up with this context in mind.

4.1. Problem description

Consider the multiplex graph $G(\Gamma, E^v)$ with Γ set of all nodes and E^v set of all edges, $v \in V$ number of layers where V is the set of drone types, K is the set of existing stations, I is the set of additional stations where $\Gamma = K \cup I$ and $K \cap I = \{\}$. M is a list of missions where each mission is defined with a single pair of pick-up and delivery stations:

$$\begin{aligned} &M = (m_1, m_2, ...m_i, ..., m_n) \\ &m_i = (\gamma_m^p, \gamma_m^d) \qquad \gamma_m \in K \end{aligned} \tag{8}$$

Edges are defined as pairs of nodes (undirected):

$$E = \{(\gamma_i, \gamma_j) \in \Gamma \mid i < j\} \tag{9}$$

$E^v \subset E$ for pairs of nodes (γ_i, γ_j) whose distance function $d(\gamma_i, \gamma_j)$ is within the range R^v of the given type of drone v:

$$E^v = \{(\gamma_i, \gamma_j) \in E \mid d(\gamma_i, \gamma_j) \leq R^v\} \qquad d(\gamma_i, \gamma_j), R^v \in \mathbb{R}^+ \tag{10}$$

A layer path is defined as a list of nodes where consecutive pairs of nodes belong to E^v for the given layer v, and which contains one pick-up node γ^p and one drop-off node γ^d, where the pick-up node must come before the drop-off node in the list:

$$\begin{aligned} &\vec{P}^v = (\gamma_1, \gamma_2, ..., \gamma_k, ..., \gamma^p, ..., \gamma^d ..., \gamma_n) \qquad \gamma \in \Gamma \\ &(\gamma_k, \gamma_{k+1}) \in E^v \end{aligned} \tag{11}$$

A mission path is a list of layer paths whose consecutive pick-up/drop-off nodes match (different layers in the multiplex graph share some or all nodes, but have different edge sets E^v), and whose first/last layer paths' pick-up/drop-off nodes match the given mission's pick-up/delivery stations:

$$\begin{aligned} &P^m = (\vec{P}_1^v, \vec{P}_2^v, ..., \vec{P}_i^v, ..., \vec{P}_n^v) \\ &\gamma^d \in \vec{P}_i^v = \gamma^p \in \vec{P}_{i+1}^v \\ &\gamma^p \in \vec{P}_1^v = \gamma_m^p \in m \in M \\ &\gamma^d \in \vec{P}_n^v = \gamma_m^d \in m \in M \end{aligned} \tag{12}$$

Similarly, a vehicle path $\vec{P}^\rho$, associated with a specific vehicle $\vec{P}^\rho \to \rho \in v \in V$, is a concatenation of layer paths $\vec{P}^v$ whose consecutive start/end nodes match:

$$\begin{aligned} &P^\rho = (\vec{P}_1^v, \vec{P}_2^v, ..., \vec{P}_i^v, ..., \vec{P}_n^v) \\ &\gamma_n \in \vec{P}_i^v = \gamma_1 \in \vec{P}_{i+1}^v \end{aligned} \tag{13}$$

4.1.1. Parameters

Type of charging infrastructure s_γ^c and drone port infrastructure s_γ^d at station γ where 0 means no infrastructure. These represent items in a given list of designs, where N_{sc} and N_{sd} are the length of each list:

$$\begin{aligned} &s_\gamma^c \in \{0, 1, ..., N_{sc}\} \\ &s_\gamma^d \in \{0, 1, ..., N_{sd}\} \end{aligned} \tag{14}$$

Number of station charge ports N_γ^c and landing ports N_γ^l:

$$\begin{aligned} &N_\gamma^c \in \mathbb{Z}^+ \\ &N_\gamma^l \in \mathbb{Z}^+ \end{aligned} \tag{15}$$

Select additional locations:

$$x_\gamma = \begin{cases} 1, & \text{if location } \gamma \text{ is active} \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

Let $\beta \subset I$ for which $x_\gamma = 0$:

$$\beta = \{\gamma \in I \,|\, x_\gamma = 0\} \tag{17}$$

Count of the number of times a given edge $e = (n_i, n_j)$ is used in a given mission m by a given vehicle type v:

$$\begin{aligned} &b_{i,j,k}^e = \begin{cases} 1, \text{if } \gamma_i = \gamma_k \in \vec{P}^v \in P^m, \text{and } \gamma_j = \gamma_{k+1} \in \vec{P}^v \in P^m \\ 1, \text{if } \gamma_j = \gamma_k \in \vec{P}^v \in P^m, \text{and } \gamma_i = \gamma_{k+1} \in \vec{P}^v \in P^m \\ 0, \text{otherwise} \end{cases} \\ &z_{evm}^e = \sum_{k=1}^{n-1} b_{i,j,k}^e \qquad e \in E^v = (\gamma_i, \gamma_j) \end{aligned} \tag{18}$$

Count of the number of times a given node γ_k is used in a given mission m by a given vehicle type v:

$$\begin{aligned} &b_{i,k}^\gamma = \begin{cases} 1, \text{if } \gamma_i = \gamma_k \in \vec{P}^v \in P^m \\ 0, \mathit{otherwise} \end{cases} \\ &z_{\gamma vm}^\gamma = \sum_{k=1}^{n} b_{i,k}^\gamma \end{aligned} \tag{19}$$

4.1.2. Constraints

The nominal (without failure events) graph is connected:

$$\forall\{\gamma_i, \gamma_j\} \in \{\Gamma - \beta\},\ \exists\, P^m \ni \forall\{\gamma_i, \gamma_j\} \tag{20}$$

Each mission m (equation 8) is completed by exactly one mission path P^m (equation 12):

$$P^m \rightarrow M \tag{21}$$

A mission cannot have the same node as its pick-up and delivery points:

$$\gamma_m^p \neq \gamma_m^d \tag{22}$$

Existing stations are always active, while additional stations can be active or inactive (equation 16):

$$\begin{aligned} x_\gamma &= 1 && \forall\gamma \in K \\ x_\gamma &\in \{0, 1\} && \forall\gamma \in I \end{aligned} \tag{23}$$

There is no infrastructure for stations in set β (equations 14-17):

$$\forall\gamma \in \beta : \begin{cases} s_\gamma^c = 0 \\ s_\gamma^d = 0 \\ N_\gamma^c = 0 \\ N_\gamma^l = 0 \end{cases} \tag{24}$$

All active stations must have charging and landing infrastructure (equations 14-17):

$$\forall\gamma \in \{\Gamma - \beta\} : \begin{cases} s_\gamma^c > 0 \\ N_\gamma^c > 0 \\ N_\gamma^l > 0 \end{cases} \tag{25}$$

The first and last nodes in a vehicle path (equation 13) must have drone port infrastructure (equation 14):

$$s_\gamma^d > 0 : \begin{cases} \forall\gamma_1 \in \vec{P}_1^v \in P^\rho \\ \forall\gamma_n \in \vec{P}_n^v \in P^\rho \\ \forall\rho \in v \\ \forall v \in V \end{cases} \tag{26}$$

The total number of drones charging at a station at any given time cannot exceed the number of charge ports:

$$d_\gamma^c \leq N_\gamma^c \tag{27}$$

The total number of drones taking off and landing at a station at any given time cannot exceed the number of landing ports (equation 15):

$$d_\gamma^t + d_\gamma^l \leq N_\gamma^l \tag{28}$$

4.2. Network Planning / Scheduling

Network planning and scheduling refers to the selection of the routes each vehicle should take, and in what order, such that the objectives of the overall optimisation are minimised.

To keep the computational cost sufficiently small for a large exploration of network designs, the following simplifications are made:

1. The flight path between each pair of nodes is precomputed once, minimising risk and time as in Figure 7 and avoiding no-fly zones. The length of the path determines whether a given drone type has enough range to fly between the nodes and how long the flight takes.
2. Deliveries have a strict priority; in this paper they are scheduled sequentially in ascending order of the distance between the pickup and drop-off locations. Missions only need consider congestion from previously scheduled deliveries.
3. Takeoff/landing/charging times are fixed for each drone type. Takeoff/landing/charging can be delayed, but not paused and resumed.
4. Each drone may only carry 1 package at a time. Since our use case involves a fixed number of pickup/delivery locations, in reality an operating network may amalgamate identical missions together up to the drone's carrying capacity.

Each mission is considered as a two-stage problem. The first stage involves selection of a drone and a route to the pickup node, followed by the second stage where a route from the pickup to the drop-off node is selected.

The algorithm may recurse during the second stage when a node shared between the two vehicle types is reached. The node can be treated as a pickup node for the other vehicle type, with the same original drop-off location. This recursion allows the package to be 'swapped' between the drone types, making use of the advantages of having a heterogeneous fleet.

The usage of each landing port and charge port is tracked, allowing for the queueing delay to be computed for a particular take-off/landing/charging attempt at a particular node and a particular time. The usage of each drone is also tracked, drones becoming available again after completing their mission. This means the edge weights in the graph search are dynamic, and the fastest route is not necessarily also the shortest route.

The algorithm used is a modified version of A-star. Node scores are initially seeded as the time that each drone is next available, at each drone port. The search heuristic is the distance from a given node to the target location, either the pickup or drop-off point. Edge weights are a dynamic property that accounts for both the nominal traversal time for the given drone type in seconds, but also the queueing delay for take-off/landing/charging, which depends on the state of the network, the node scores, and also the path taken so far in the case of back-tracking from a pickup point.

4.3. Network Design

At a strategic level, the goal is to calculate a set of optimal network design configurations, in a Pareto sense, for a network of drones delivering medical supplies between NHS institutions within Scotland.

4.3.1. Problem statement

Given a list of:

1. Locations
   (a) Coordinates
   (b) Type: source, receiver, source-receiver, intermediate
2. Infrastructure types
   (a) Charging ports
   (b) Landing ports
   (c) Drone storage ports
3. Vehicle types
4. Missions
5. Metrics

Select:

1. Locations to be included in the network
2. For each location, the list of attached infrastructures
3. Vehicles
   (a) Type and number
   (b) Where they are stored
4. List of routing plans (drone type, list of nodes)

Such that:

1. Time
2. Cost
3. Risk

Are minimised.

4.3.2. Fixed parameters

The existing airports A and NHS locations H consist of the fixed set K (Figure 1):

$$\begin{aligned} &H \subset K \\ &A \subset K \\ &H \cup A = K \\ &H \cap A = \{\} \end{aligned} \tag{29}$$

Scotland is divided into fourteen health boards (Figure 9):

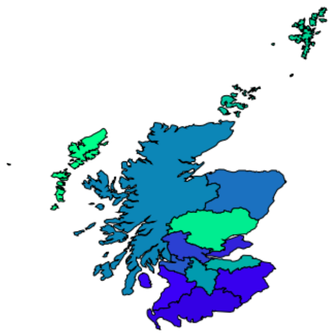

Figure 9: NHS health board demarcations.

Each of the NHS locations and airports are associated with a health board $\theta \in \Theta$ based on these boundaries. Then, at least one location within each health board is selected as a 'hub' Ω, at least one of which must be an NHS location:

$$\begin{aligned} &\gamma \in \theta \\ &\theta_1 \cup \theta_2 \cup ... \cup \theta_{14} = \Gamma \\ &\{\theta_i \cap \theta_j : i \neq j\} = \{\} \\ &\forall \theta \in \Theta : \begin{cases} \Omega_\theta \neq \{\} \\ \Omega_\theta \cap H \neq \{\} \end{cases} \end{aligned} \tag{30}$$

Hubs are treated as source-receivers, whereas all other locations in K are treated as receivers. All additional stations I are treated as intermediate, meaning they cannot send or receive deliveries, and only act as a charging port to optimally connect the network considering the maximum range of the drones.

For the purpose of network design for Scotland, a representative set of missions is generated at the health board level, where missions are planned from every source-receiver (hub) to every receiver and every other source-receiver:

$$(\gamma^p_{i\theta}, \gamma^d_{j\theta}) \in M_\theta : \begin{cases} \forall i \in \Omega_\theta \\ \forall j \in \{\{\theta - \Omega_\theta\} \cap K\} \\ i \neq j \end{cases} \tag{31}$$

Additional missions are generated at the national level between every pair of hubs across all boards:

$$(\gamma^p_{i\Gamma}, \gamma^d_{j\Gamma}) \in M_\Gamma : \begin{cases} \forall i \in \Omega_\theta \\ \forall j \in \Omega_\theta \\ \forall \theta \in \Theta \\ i \neq j \end{cases} \tag{32}$$

Two types of drone are considered, a smaller vertical take-off and landing (VTOL) drone (type 1), and a larger fixed-wing drone (type 2). They are each assigned an assumed nominal range, speed, and battery swap time (table 1).

Vehicle	R^v	$t_{\gamma c}$	$t_{\gamma t}$	$t_{\gamma l}$	Υ
v_1	70 km	300 s	60 s	300 s	32 m/s
v_2	700 km	1800 s	300 s	600 s	69 m/s

Table 1: Nominal performance for each drone type. R^v comes from equation 10; $t_{\gamma c}$ represents the time taken to swap batteries; $t_{\gamma t}$ represents the time taken to take off; $t_{\gamma l}$ represents the time taken to land; Υ represents the nominal velocity.

Four types of charge/drone port infrastructure are considered, including the null type for no infrastructure (table 2):

Charge port type	Drone type	c_s^c	$N^c = N^l$
S_0^c	N/A	0	0
S_1^c	1	1	2
S_2^c	1	2	4
S_3^c	1	3	6
S_4^c	2	1	1
Drone port type	**Drone type**	c_s^d	N_{dn}
S_0^d	N/A	0	0
S_1^d	1	1	2
S_2^d	1	2	4
S_3^d	1	3	6
S_4^d	2	2	1

Table 2: Nominal performance/cost for each station infrastructure. c_s^c is the cost of the charge station type; c_s^d is the cost of the drone station type; N^c is the number of charge ports; N^l is the number of landing ports; N_{dn} is the number of drones that can be stored.

The network is seeded with the set of additional stations I based on the maximum range R^v of the VTOL drone v_1, as a mix of a grid pattern and manually selected locations, as shown in Figure 10:

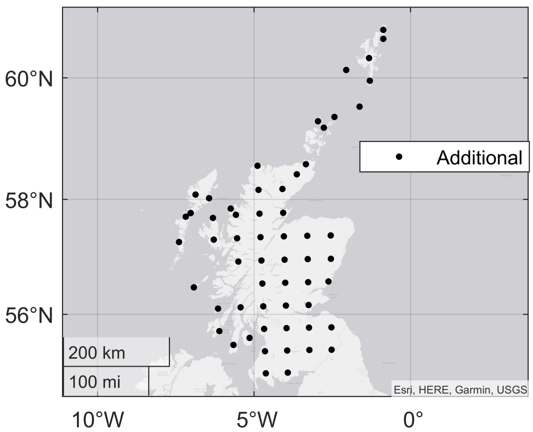

Figure 10: Potential additional drone stations within Scotland.

4.3.3. *Search parameters*

To solve the stated problem, the following optimisation variables are selected $\mathbb{X} = \{X, S^c, S^d, \mathbf{P}\}$:

$$\begin{aligned} x_\gamma &\in X && \forall \gamma \in I \\ s_\gamma^c &\in S^c && \forall \gamma \in \Gamma \\ s_\gamma^d &\in S^d && \forall \gamma \in \Gamma \\ P^m &\in \mathbf{P} && \forall m \in M \end{aligned}$$

where x_γ is from (16) and represents the selection process of new stations; s_γ^c and s_γ^d are from (14) and represent the selection of charge and drone port infrastructure; P^m is from (12) and represents the selection of a path for each mission.

4.3.4. *Objectives*

The network design is a multi-objective optimisation problem for which the vector of objectives can be represented as:

$$\mathbf{f}(\mathbf{x}) = [f_1, f_2, ..., f_m]^T. \tag{33}$$

For a robust solution, uncertainty needs to be propagated and quantified specific metrics (e.g., network reliability and resilience):

$$\text{Resilience} = \sum_{\lambda \in \Lambda} p_\lambda \Phi_\lambda(\mathbf{d}, \mathbf{u}_\lambda) \tag{1 revisited}$$

The delivery time and capital cost expenditure are also used as performance indicator:

$$\text{Time} = \sum_{\substack{m \in M, \\ e \in E^v \\ \gamma \in \Gamma \\ v \in V}} \mathbf{E}(t_{ev} z_{evm}^e + t_{\gamma v} z_{\gamma v m}^\gamma) \tag{34}$$

Where t_{ev} is the time taken for vehicle v to travel edge e, and $t_{\gamma v}$ is the time taken for the same vehicle to recharge or swap batteries at node γ (loading/unloading can be done simultaneously and is assumed to take less time).

$$\text{Cost} = \sum_{\gamma \in \Gamma} \mathbf{E}(c_{\gamma s}^c + c_{\gamma s}^d) \tag{35}$$

Where $c_{\gamma s}^c$ and $c_{\gamma s}^d$ are the costs associated with charge/drone port type s_γ^c and s_γ^d (equation 14), respectively:

$$\begin{aligned} \{0, 1, ..., N_{sc}\} &\rightarrow c_s^c \ni c_{\gamma s}^c \\ \{0, 1, ..., N_{sd}\} &\rightarrow c_s^d \ni c_{\gamma s}^d \end{aligned} \tag{36}$$

Other performance indicators and metrics can also be included explicitly in the optimisation problem (e.g. maximum flow).

4.3.5. *Incremental constraints*

1. Hubs must have drone port infrastructure:
$$s_\gamma^d > 0 : \begin{cases} \forall \gamma \in \Omega_\theta \\ \forall \theta \in \Theta \end{cases}$$
2. The fixed-wing drone v_2 can only visit airfield nodes $A \subset K$ (equation 29)
3. No drone can take-off or land at the same time and at the same node as a fixed-wing drone v_2 is taking off or landing
4. The number of initial drones of a given type at each node is equal to the maximum storage capacity of that type at that node.

5. Results

The aim is to find a Pareto front of optimal network designs to demonstrate trade-offs and identify and learn from patterns in the optimal network designs that correspond to these trade-offs. The design space and objective functions are outlined in sections 4.3.3 and 4.3.4, respectively.

For this study, the average resilience across 10 failure cases is considered, in which each node except the hub nodes Ω has a 20% failure probability. To avoid a stochastic optimisation, the failure cases are kept constant throughout.

A multi-objective genetic algorithm is used, minimising capital/time cost, and maximising resilience. The population size is set to 25 over 250 generations, resulting in 6250 function evaluations. With 10 failure cases and 1 nominal case considered per function evaluation, this resulted in 68,750 network design evaluations in total.

The Pareto front for the three objectives is constructed by storing the result of all function evaluations and plotting all design costs and resilience across the whole optimisation, excluding points with a strictly superior alternative, as shown in Figure 11:

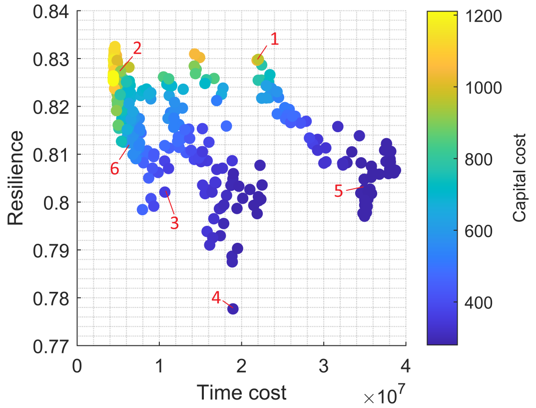

Figure 11: Pareto front of network design solutions in time cost, capital cost, and resilience. Some points are annotated for reference.

Each network design may be plotted as points in physical (latitude-longitude) space, connected with edges corresponding only to the nominal planned missions. These plots are hereafter referred to as *constellation plots*, an example of which is shown for network design point 6 of Figure 11 in Figure 12. Square markers represent charge port infrastructure, whereas crosses represent drone port infrastructure, with the square marker always shown on top. Colours green, blue, red, and orange correspond to infrastructure types 1, 2, 3, and 4, respectively, from table 1. Black edges represent flights with the smaller VTOL drone v_1, and red edges represent flights with the larger fixed-wing drone v_2. For simplicity, straight lines are shown, actual paths taken are pre-computed accounting for no-fly zones.

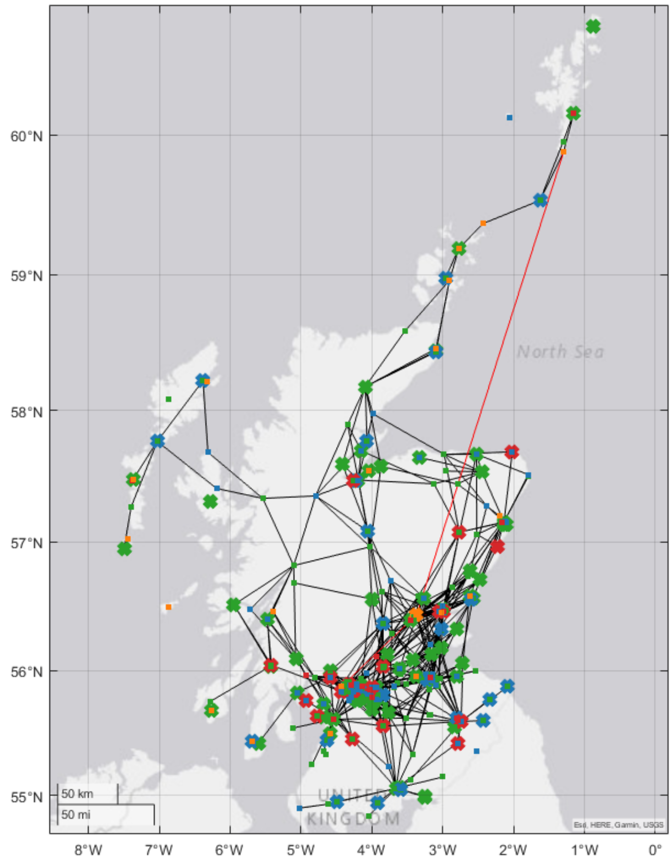

Figure 12: Constellation plot for network design Pareto point 6 (Fig. 11) showing infrastructure and nominal mission routes. A description of markers and edges is provided earlier in this section.

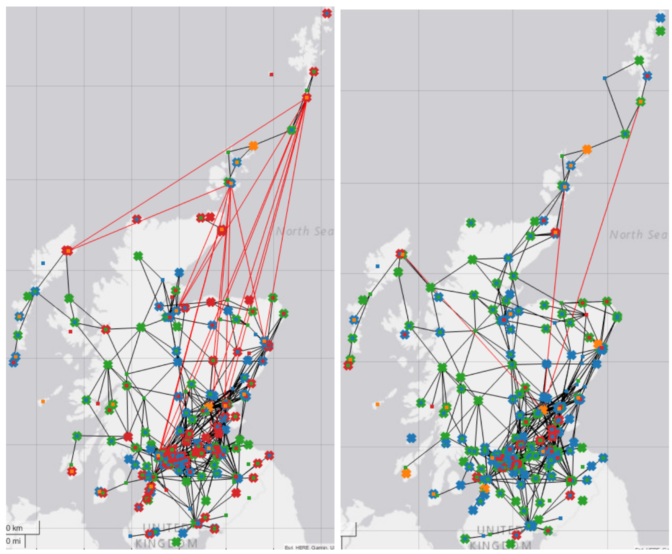

Figure 13: Constellation plot for network design Pareto points 1 (left) and 2 (right) (Fig. 11) showing infrastructure and nominal mission routes. A description of markers and edges is provided earlier in this section.

Comparing points 1 and 2 in Figure 11, it can be seen that a large increase in delivery time and a moderate increase in capital cost only trade off for a marginal increase in resilience.

The constellation plots for these networks are shown in Figure 13. It can be seen that network 1 relies much more heavily on the fixed-wing drone for deliveries to the northern isles (Orkney and Shetland), being disconnected from the main VTOL network otherwise. Fixed-wing flight deliveries results in a slower network, with fewer drones available to perform these deliveries for the same capital cost, and longer take-off/landing/charge times. Although, there are a sufficient number of alternate airfields available, making the network resilient to failure. The more centralised design also minimises disruption for the majority of the deliveries.

Conversely, network 2 (Fig. 13) has a more decentralised structure, relying on a fully connected VTOL network layer for a slightly less resilient but much faster network.

Networks 3 and 5 (Figure 14, below) represent smaller capital cost, but slower and less resilient alternatives to networks 1 and 2. It is clear that there are many fewer drone ports in the networks, leading to fewer redundancies, and in the case of node failures, causing delays where each drone has to complete more missions. Similar to networks 1 and 2, the network relying more heavily on fixed-wing drone flights is much slower for relatively marginal gains in resilience and reductions in capital cost.

Finally, the least resilient network on the Pareto front, point 4 (Figure 15, below). The main differentiator for this network among the others is a concentration down to only a few larger drone ports whilst maintaining connectivity to the mainland north, giving the network a faster performance than network 1 (fig 11) at a vastly reduced cost. The smaller number of available drones, however, leads to a much smaller capacity to operate efficiently during failure cases, where each drone is given a much greater burden of deliveries, especially when one or more of the major (red) drone ports fails.

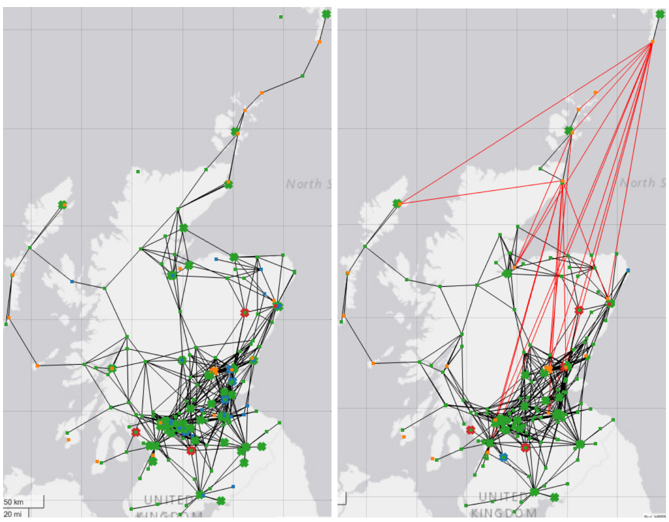

Figure 14: Constellation plot for network design Pareto points 3 (left) and 5 (right) (Fig. 11) showing infrastructure and nominal mission routes. A description of markers and edges is provided earlier in this section.

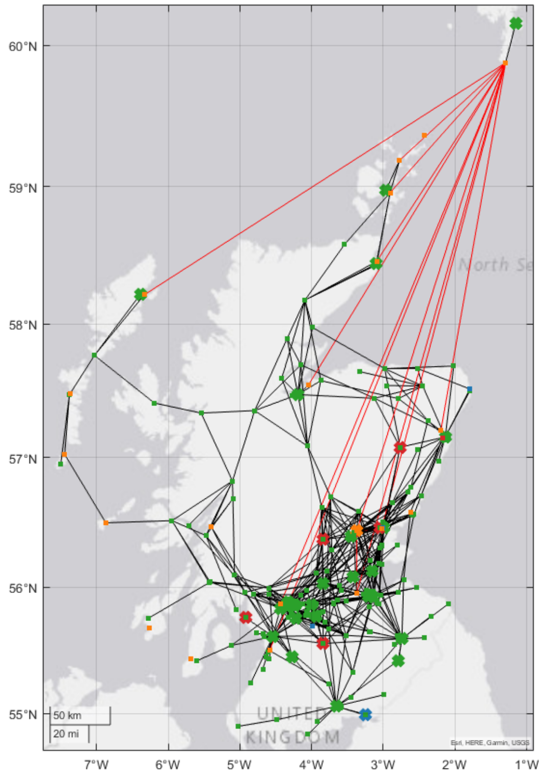

Figure 15: Constellation plot for network design Pareto point 4 (Fig. 11) showing infrastructure and nominal mission routes. A description of markers and edges is provided earlier in this section.

6. Conclusions

A multi-objective optimisation is defined and executed for the design of a medical drone logistical network across Scotland. A mix of two drone types are considered, a smaller vertical take-off and landing (VTOL) drone, which can operate between drone ports built on arbitrary (on land) locations, especially labs and hospitals where medical goods are delivered from/to; and a larger fixed-wing drone which may only operate between established airfields/airports.

The three optimisation objectives are: capital cost, related to the investment required for the given station infrastructure; time cost, related to the number of deliveries the network can complete in a given time-frame; and resilience, which is a measure of the network's ability to recover from failures and maximise efficiency with respect to the nominal (no-failures) condition.

Out of the networks on the Pareto front, it is shown that networks which rely more on the fixed-wing drone are much slower and only marginally less costly than their counterparts with similar levels of resilience.

There is a very strong link between resilience and capital cost, which is mainly a consequence of the availability of redundant drones to complete deliveries instead of increasing the burden on existing non-failed drones.

Overall, the developed methodology and analysis is an effective tool for the conceptual design of drone logistics networks. Applied to Scotland, a Pareto front of designs that could be implemented is demonstrated. In future work, the network operation will be investigated, bringing in a greater level of detail into the drone simulation process, and including a dynamic network response to failures.

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Data Availability

The data that support the findings of this study are available from the corresponding author, Bryn Jones, upon reasonable request.

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